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Faculté des sciences et des technologies (FST)

Administration Système_ TD N° 6

Préparé par : Brian ANTOINE

Présenté à: M. Ismaël SAINT AMOUR

Niveau : L3

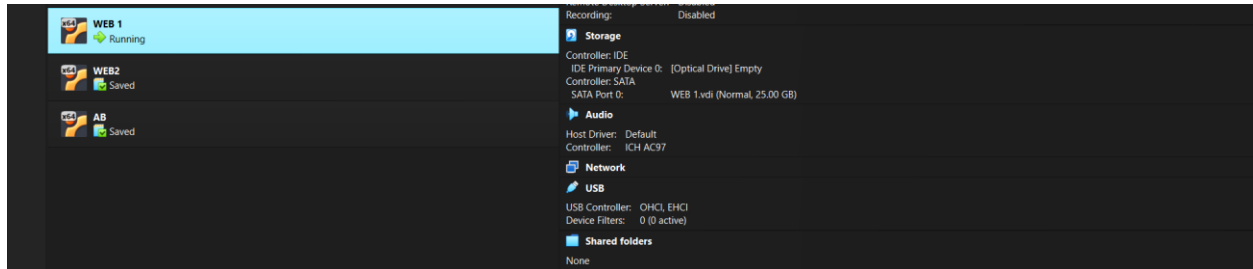
Date: 12/04/2026

L'objectif de ce TD est de :

1. Déployer une architecture web résiliente
2. Configurer un site dynamique PHP
3. Optimiser les performances (cache)
4. Sécuriser avec HTTPS
5. Tester la montée en charge

E-commerce HA + Paiement simulé

➤ Machines



1. Installation de Nginx + PHP sur Web1

```
web1@WEB1: ~  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
  
web1@WEB1:~$ sudo apt update  
[sudo] password for web1:  
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease  
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease  
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease  
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
30 packages can be upgraded. Run 'apt list --upgradable' to see them.  
web1@WEB1:~$ sudo apt install nginx php-fpm -y  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
The following additional packages will be installed:  
  nginx-common php-common php8.3-cli php8.3-common php8.3-fpm php8.3-opcache php8.3-readline  
Suggested packages:  
  fcgiwrap nginx-doc ssl-cert php-pear  
The following NEW packages will be installed:  
  nginx nginx-common php-common php-fpm php8.3-cli php8.3-common php8.3-fpm php8.3-opcache  
  php8.3-readline  
0 upgraded, 9 newly installed, 0 to remove and 30 not upgraded.  
Need to get 5,549 kB of archives.  
After this operation, 24.1 MB of additional disk space will be used.  
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 nginx-common all 1.24.0-2ubuntu7.
```

```

web1@WEB1:~$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-04-13 14:16:20 UTC; 2min 15s ago
     Docs: man:nginx(8)
  Main PID: 6171 (nginx)
    Tasks: 2 (limit: 2263)
   Memory: 1.7M (peak: 3.7M)
      CPU: 66ms
   CGroup: /system.slice/nginx.service
           └─6171 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
              └─6174 "nginx: worker process"

Apr 13 14:16:20 WEB1 systemd[1]: Starting nginx.service - A high performance web server and a re>
Apr 13 14:16:20 WEB1 systemd[1]: Started nginx.service - A high performance web server and a rev>
lines 1-14/14 (END)
[1]+  Stopped                  sudo systemctl status nginx
web1@WEB1:~$

```

```

web1@WEB1:~$ systemctl list-unit-files | grep -i php
php8.3-fpm.service          enabled          enabled
phpsessionclean.service    static          -
phpsessionclean.timer       enabled          enabled
web1@WEB1:~$

```

➤ Création du site e-commerce sur Web1

```

web1@WEB1: ~
No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
web1@WEB1:~$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-04-13 14:16:20 UTC; 2min 15s ago
     Docs: man:nginx(8)
  Main PID: 6171 (nginx)
    Tasks: 2 (limit: 2263)
   Memory: 1.7M (peak: 3.7M)
      CPU: 66ms
   CGroup: /system.slice/nginx.service
           └─6171 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
              └─6174 "nginx: worker process"

Apr 13 14:16:20 WEB1 systemd[1]: Starting nginx.service - A high performance web server and a re>
Apr 13 14:16:20 WEB1 systemd[1]: Started nginx.service - A high performance web server and a rev>
lines 1-14/14 (END)
[1]+  Stopped                  sudo systemctl status nginx
web1@WEB1:~$ systemctl list-unit-files | grep -i php
php8.3-fpm.service          enabled          enabled
phpsessionclean.service    static          -
phpsessionclean.timer       enabled          enabled
web1@WEB1:~$ sudo mkdir -p /var/www/shop
web1@WEB1:~$ sudo nano /var/www/shop/index.php
web1@WEB1:~$

```

➤ Permissions

```

web1@WEB1:~$ sudo nano /var/www/shop/index.php
web1@WEB1:~$ sudo chown -R www-data:www-data /var/www/
web1@WEB1:~$ sudo chmod -R 755 /var/www/
web1@WEB1:~$

```

➤ Configuration de Nginx : création du virtual host

```
web1@WEB1: ~  
web1@WEB1:~$ sudo chown -R www-data:www-data /var/www/  
web1@WEB1:~$ sudo chmod -R 755 /var/www/  
web1@WEB1:~$ sudo nano /etc/nginx/sites-available/shop  
web1@WEB1:~$ cat /etc/nginx/sites-aivalable/shop  
cat: /etc/nginx/sites-aivalable/shop: No such file or directory  
web1@WEB1:~$ cat /etc/nginx/sites-available/shop  
server {  
    listen 80;  
    server_name shop.devcloud.local;  
  
    root /var/www/shop;  
    index index.php;  
  
    location / {  
        try_files $uri $uri/ =404;  
    }  
  
    location ~ \.php$ {  
        include snippets/fastcgi-php.conf;  
        fastcgi_pass unix:/run/php/php8.2-fpm.sock;  
    }  
}  
}~server {  
    listen 80;  
    server_name shop.devcloud.local;  
  
    root /var/www/shop;  
    index index.php;  
  
    location / {  
        try_files $uri $uri/ =404;
```

➤ Activer le site

```
web1@WEB1: ~  
    location ~ \.php$ {  
        include snippets/fastcgi-php.conf;  
        fastcgi_pass unix:/run/php/php8.2-fpm.sock;  
    }  
}  
EOF  
web1@WEB1:~$ cat /etc/nginx/sites-available/shop  
server {  
    listen 80;  
    server_name shop.devcloud.local;  
  
    root /var/www/shop;  
    index index.php;  
  
    location / {  
        try_files $uri $uri/ =404;  
    }  
  
    location ~ \.php$ {  
        include snippets/fastcgi-php.conf;  
        fastcgi_pass unix:/run/php/php8.2-fpm.sock;  
    }  
}  
web1@WEB1:~$ sudo ln -s /etc/nginx/sites-available/shop /etc/nginx/sites-enabled/  
web1@WEB1:~$ sudo nginx -t  
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok  
nginx: configuration file /etc/nginx/nginx.conf test is successful  
web1@WEB1:~$ sudo systemctl restart nginx  
web1@WEB1:~$
```

2. Installation de Nginx + PHP sur Web2

```
web2@WEB2:~$ sudo apt update
[sudo] password for web2:
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Get:5 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1,892 kB]
Get:6 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1,584 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [254 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble-updates/main Translation-en [344 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1,170 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [2,934 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [681 kB]
Get:12 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1,666 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [323 kB]
Get:14 http://archive.ubuntu.com/ubuntu noble-updates/multiverse Translation-en [7,752 B]
Get:15 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [225 kB]
Fetched 11.3 MB in 2min 6s (89.7 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
35 packages can be upgraded. Run 'apt list --upgradable' to see them.
web2@WEB2:~$ sudo apt install nginx php-fpm -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  nginx-common php-common php8.3-cli php8.3-common php8.3-fpm php8.3-opcache php8.3-readline
Suggested packages:
```

```
web2@WEB2:~$ systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-04-13 15:13:42 UTC; 15min ago
     Docs: man:nginx(8)
    Main PID: 6728 (nginx)
      Tasks: 3 (Limit: 4601)
    Memory: 2.4M (peak: 5.1M)
       CPU: 63ms
    CGroup: /system.slice/nginx.service
            └─6728 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
              └─6731 "nginx: worker process"
                └─6732 "nginx: worker process"

web2@WEB2:~$ systemctl list-unit-files | grep -i php
php8.3-fpm.service             enabled          enabled
phpsessionclean.service       static          -
phpsessionclean.timer         enabled         enabled

web2@WEB2:~$ systemctl status php8.2-fpm
Unit php8.2-fpm.service could not be found.
web2@WEB2:~$ systemctl status php8.3-fpm
● php8.3-fpm.service - The PHP 8.3 FastCGI Process Manager
   Loaded: loaded (/usr/lib/systemd/system/php8.3-fpm.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-04-13 15:14:16 UTC; 17min ago
     Docs: man:php-fpm8.3(8)
  Process: 8299 ExecStartPost=/usr/lib/php/php-fpm-socket-helper install /run/php/php-fpm.sock /etc/php/8.3/fpm/pool...
    Main PID: 8296 (php-fpm8.3)
    Status: "Processes active: 0, idle: 2, Requests: 0, slow: 0, Traffic: 0req/sec"
      Tasks: 3 (Limit: 4601)
    Memory: 7.5M (peak: 8.1M)
       CPU: 284ms
```

➤ Création du site e-commerce sur Web2

+

```
web2@WEB2: ~  
php8.3-fpm.service          enabled          enabled  
phpsessionclean.service    static          -  
phpsessionclean.timer      enabled         enabled  
web2@WEB2:~$ systemctl status php8.2-fpm  
Unit php8.2-fpm.service could not be found.  
web2@WEB2:~$ systemctl status php8.3-fpm  
● php8.3-fpm.service - The PHP 8.3 FastCGI Process Manager  
   Loaded: loaded (/usr/lib/systemd/system/php8.3-fpm.service; enabled; preset: enabled)  
   Active: active (running) since Mon 2026-04-13 15:14:16 UTC; 17min ago  
     Docs: man:php-fpm8.3(8)  
  Process: 8299 ExecStartPost=/usr/lib/php/php-fpm-socket-helper install /run/php/php-fpm.sock /etc/php/8.3/fpm/pool.  
 Main PID: 8296 (php-fpm8.3)  
    Status: "Processes active: 0, idle: 2, Requests: 0, slow: 0, Traffic: 0req/sec"  
      Tasks: 3 (limit: 4601)  
    Memory: 7.5M (peak: 8.1M)  
       CPU: 284ms  
    CGroup: /system.slice/php8.3-fpm.service  
            └─8296 "php-fpm: master process (/etc/php/8.3/fpm/php-fpm.conf)"  
              └─8297 "php-fpm: pool www"  
                └─8298 "php-fpm: pool www"  
[1]+  Stopped                  systemctl status php8.3-fpm  
web2@WEB2:~$ udo mkdir -p /var/www/shop  
Command 'udo' not found, but can be installed with:  
sudo apt install udo  
web2@WEB2:~$ sudo mkdir -p /var/www/shop  
[sudo] password for web2:  
web2@WEB2:~$ sudo mkdir -p /var/www/shop  
web2@WEB2:~$ sudo nano /var/www/shop/index.php  
web2@WEB2:~$
```

➤ Permissions

```
web2@WEB2:~$ sudo chown -R www-data:www-data /var/www/  
web2@WEB2:~$ sudo chmod -R 755 /var/www/  
web2@WEB2:~$
```

➤ Configuration de Nginx : création du virtual host

```
web2@WEB2:~$ sudo chown -R www-data:www-data /var/www/  
[sudo] password for web2:  
web2@WEB2:~$ sudo chmod -R 755 /var/www/  
web2@WEB2:~$ sudo nano /etc/nginx/sites-available/shop  
web2@WEB2:~$ sudo ln -s /etc/nginx/sites-available/shop /etc/nginx/sites-enabled/  
ln: failed to create symbolic link '/etc/nginx/sites-enabled/shop': File exists  
web2@WEB2:~$ sudo rm -f /etc/nginx/sites-enabled/default  
web2@WEB2:~$ sudo nginx -t  
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok  
nginx: configuration file /etc/nginx/nginx.conf test is successful  
web2@WEB2:~$ sudo systemctl restart nginx  
web2@WEB2:~$
```

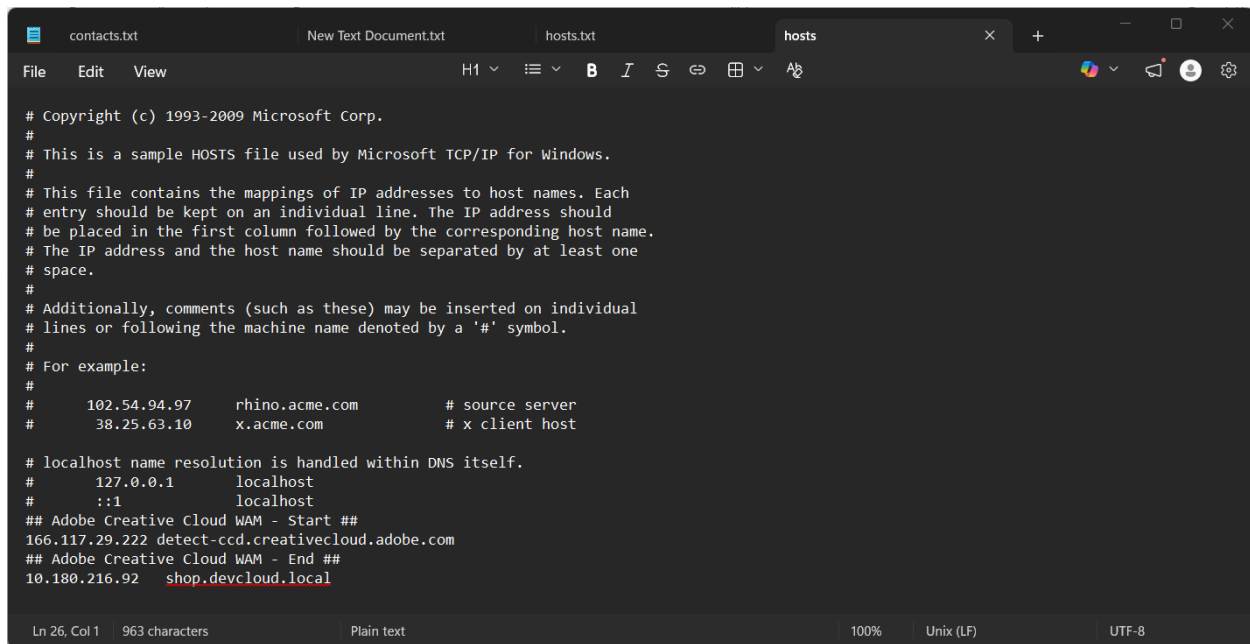
3. Configuration de Load Balancer

➤ J'ai pris mon serveur initial comme load balancer

```
bribri@bribri: ~  
bribri@10.180.216.92's password:  
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-107-generic x86_64)  
  
* Documentation:  https://help.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:        https://ubuntu.com/pro  
  
System information as of Mon Apr 13 10:13:56 PM UTC 2026  
  
System load:  0.03          Processes:            129  
Usage of /:   15.0% of 24.44GB Users logged in:        1  
Memory usage: 7%          IPv4 address for enp0s3: 10.180.216.92  
Swap usage:   0%  
  
* Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s  
  just raised the bar for easy, resilient and secure K8s cluster deployment.  
  
https://ubuntu.com/engage/secure-kubernetes-at-the-edge  
  
Expanded Security Maintenance for Applications is not enabled.  
  
16 updates can be applied immediately.  
To see these additional updates run: apt list --upgradable  
  
1 additional security update can be applied with ESM Apps.  
Learn more about enabling ESM Apps service at https://ubuntu.com/esm  
  
Last login: Mon Apr 13 17:49:29 2026 from 192.168.2.123  
bribri@bribri:~$
```

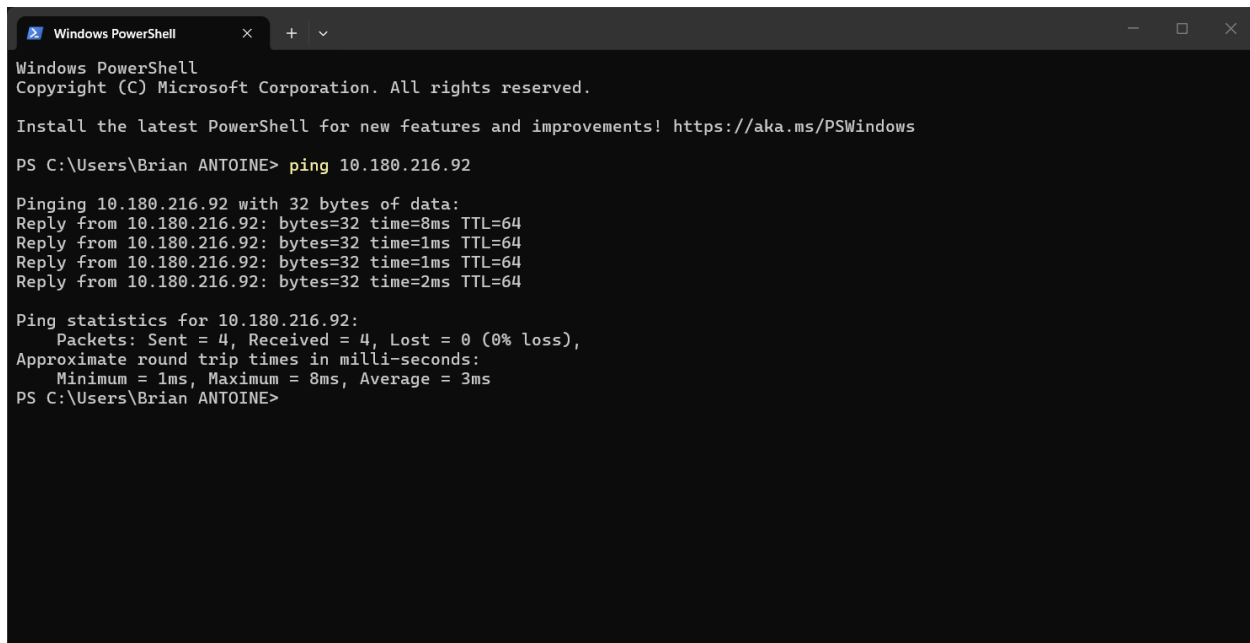
```
bribri@bribri: ~  
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease  
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease  
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
21 packages can be upgraded. Run 'apt list --upgradable' to see them.  
bribri@bribri:~$ sudo apt install nginx -y  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
nginx is already the newest version (1.24.0-2ubuntu7.6).  
  
bribri@bribri:~$ sudo nginx -t  
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok  
nginx: configuration file /etc/nginx/nginx.conf test is successful  
bribri@bribri:~$ sudo systemctl restart nginx  
bribri@bribri:~$ sudo systemctl status nginx  
● nginx.service - A high performance web server and a reverse proxy server  
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)  
   Active: active (running) since Mon 2026-04-13 17:24:01 UTC; 15s ago  
     Docs: man:nginx(8)  
  Process: 1997 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)  
  Process: 1999 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)  
 Main PID: 2001 (nginx)  
    Tasks: 2 (limit: 4601)  
  Memory: 1.5M (peak: 1.9M)  
     CPU: 35ms  
    CGroup: /system.slice/nginx.service
```


4. Tests depuis Windows



The screenshot shows a text editor window with the file 'hosts.txt' open. The editor has a dark theme and a menu bar with 'File', 'Edit', and 'View'. The status bar at the bottom indicates 'Ln 26, Col 1', '963 characters', 'Plain text', '100%', 'Unix (LF)', and 'UTF-8'. The content of the file is as follows:

```
# Copyright (c) 1993-2009 Microsoft Corp.
#
# This is a sample HOSTS file used by Microsoft TCP/IP for Windows.
#
# This file contains the mappings of IP addresses to host names. Each
# entry should be kept on an individual line. The IP address should
# be placed in the first column followed by the corresponding host name.
# The IP address and the host name should be separated by at least one
# space.
#
# Additionally, comments (such as these) may be inserted on individual
# lines or following the machine name denoted by a '#' symbol.
#
# For example:
#
#       102.54.94.97       rhino.acme.com       # source server
#       38.25.63.10       x.acme.com          # x client host
#
# localhost name resolution is handled within DNS itself.
#       127.0.0.1         localhost
#       ::1               localhost
## Adobe Creative Cloud WAM - Start ##
166.117.29.222 detect-ccd.creativecloud.adobe.com
## Adobe Creative Cloud WAM - End ##
10.180.216.92  shop.devcloud.local
```



The screenshot shows a Windows PowerShell terminal window. The title bar says 'Windows PowerShell'. The output of the terminal is as follows:

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Brian ANTOINE> ping 10.180.216.92

Pinging 10.180.216.92 with 32 bytes of data:
Reply from 10.180.216.92: bytes=32 time=8ms TTL=64
Reply from 10.180.216.92: bytes=32 time=1ms TTL=64
Reply from 10.180.216.92: bytes=32 time=1ms TTL=64
Reply from 10.180.216.92: bytes=32 time=2ms TTL=64

Ping statistics for 10.180.216.92:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 8ms, Average = 3ms
PS C:\Users\Brian ANTOINE>
```

DevClo XDevClo XDevClo X

Not secure10.18...Summarizeaあ☆☆=

Chat



DevCloud Shop

Serveur : WEB1 | Session : 2I7ptbdoj8h29ub0dpbv1k8u1



Catalogue produits

Produit	Prix	Action
Laptop	800 €	+ Ajouter
Phone	500 €	+ Ajouter
Headphones	100 €	+ Ajouter
Tablet	350 €	+ Ajouter



Panier

Produit	Quantité	Sous-total	Action
Phone	1	500 €	Retirer
Headphones	1	100 €	Retirer
Total		600 €	

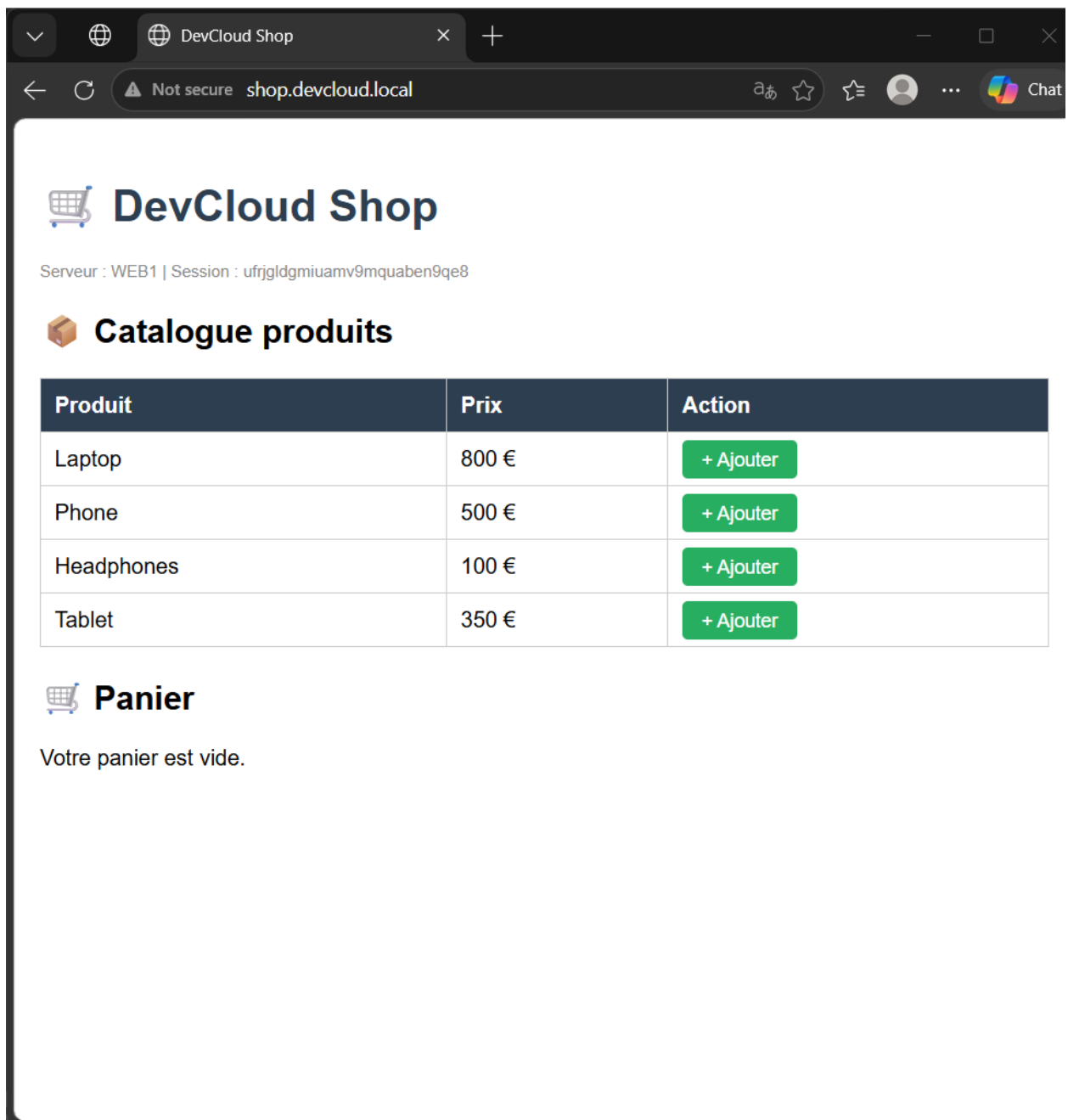


Valider la commande

Total à payer : 600 €

 Payer maintenant (simulation)





DevCloud Shop

DevCloud Shop

+

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□

×

←↻⚠ Not secure shop.devcloud.local/index.php🔍☆👤⋮🗯 Chat

 **DevCloud Shop**

Serveur : WEB1 | Session : ufrjgldgmiamv9mquaben9qe8

 **Catalogue produits**

Produit	Prix	Action
Laptop	800 €	+ Ajouter
Phone	500 €	+ Ajouter
Headphones	100 €	+ Ajouter
Tablet	350 €	+ Ajouter

 **Panier**

Produit	Quantité	Sous-total	Action
Laptop	1	800 €	Retirer
Tablet	1	350 €	Retirer
Total		1150 €	

 **Valider la commande**

Total à payer : 1150 €

 Payer maintenant (simulation)

Conclusion

Ce TD m'a permis de comprendre concrètement le fonctionnement d'une architecture web haute disponibilité, que j'avais jusqu'ici uniquement vue en théorie. En déployant moi-même les trois machines virtuelles (Load Balancer(Bribri1), WEB1 et WEB2), j'ai pu appréhender les rôles distincts de chaque composant et la façon dont ils interagissent ensemble pour garantir la continuité de service.